

Visual Storytelling of Big Imaging Data

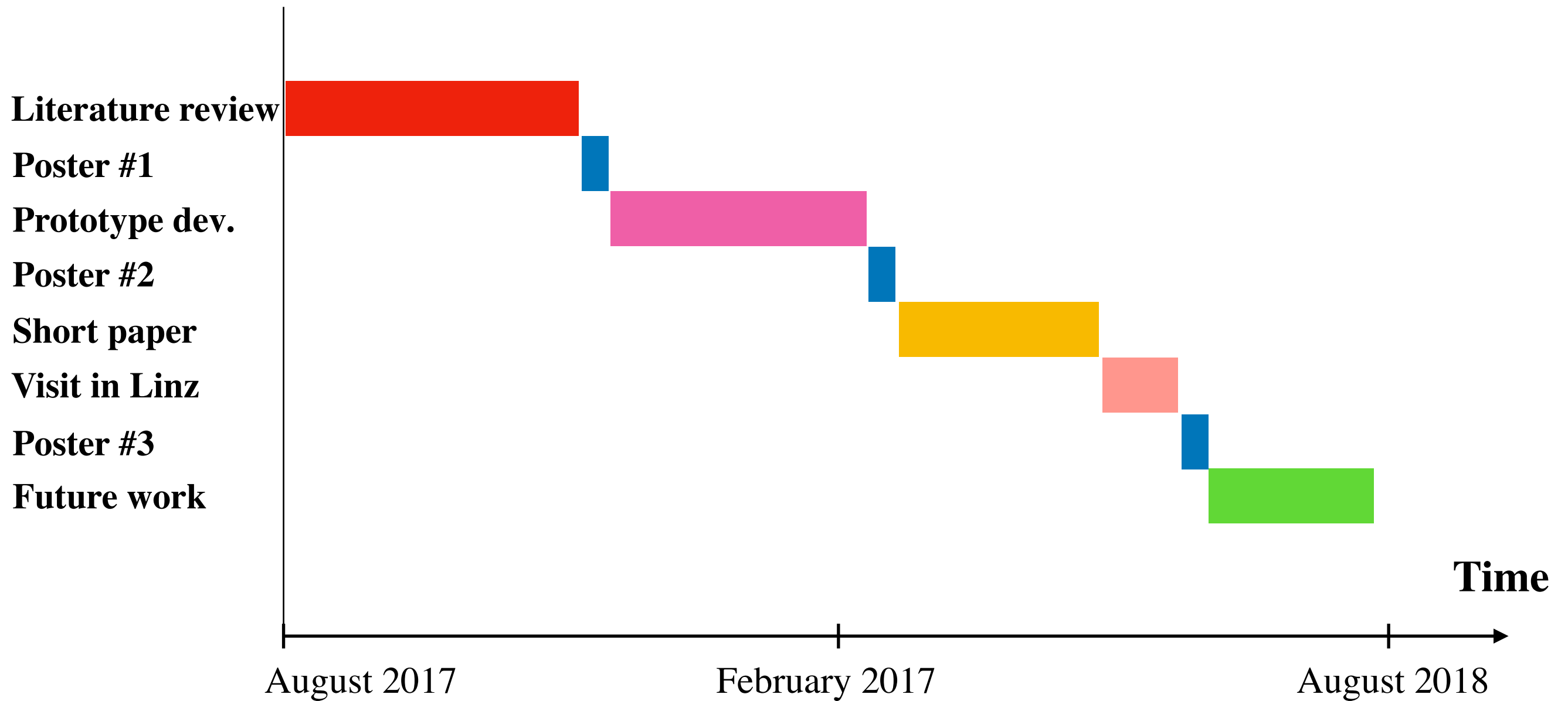
First year

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Literature review and radiology workflow survey

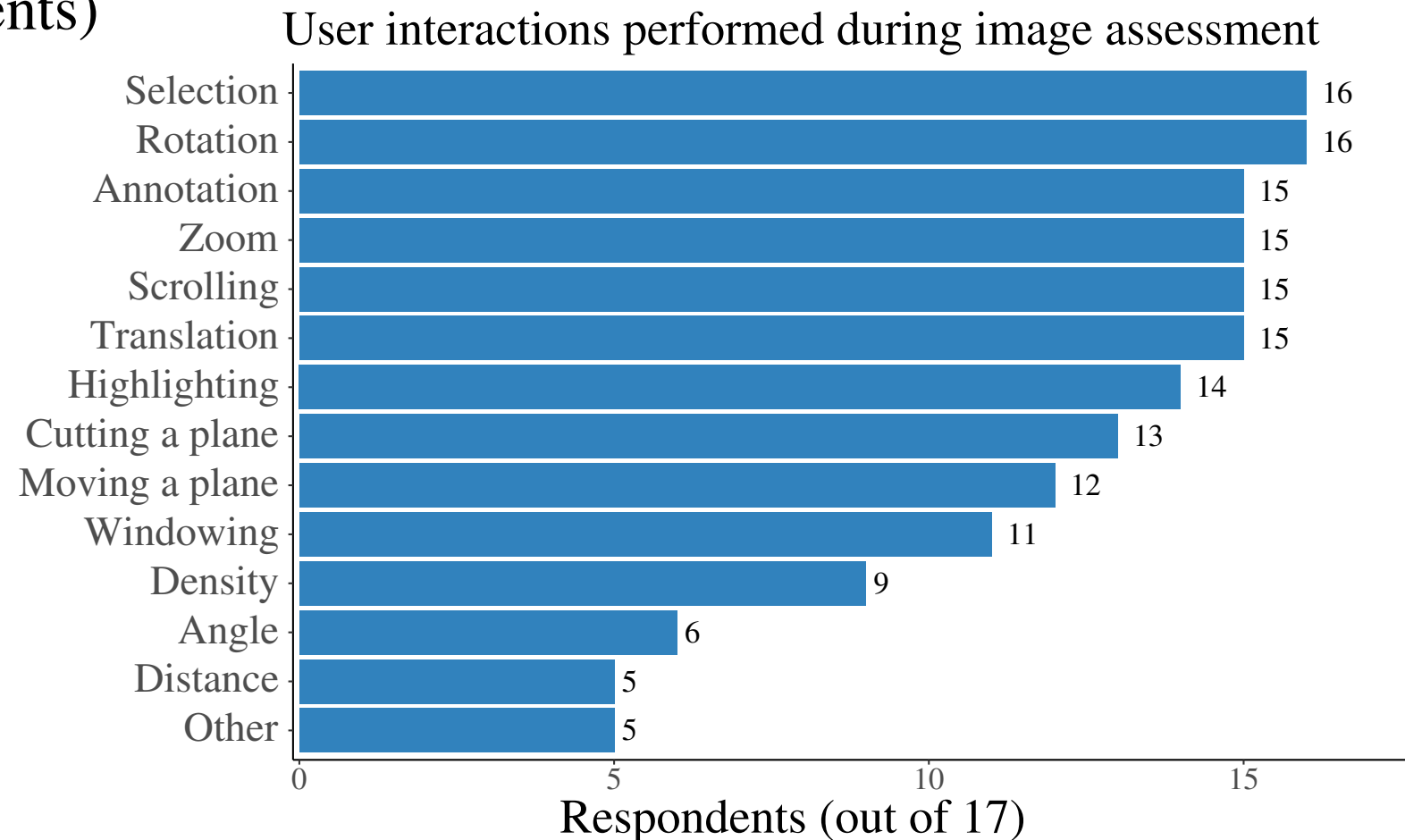
17 participants (i.e., radiologists, residents)

15 questions about:

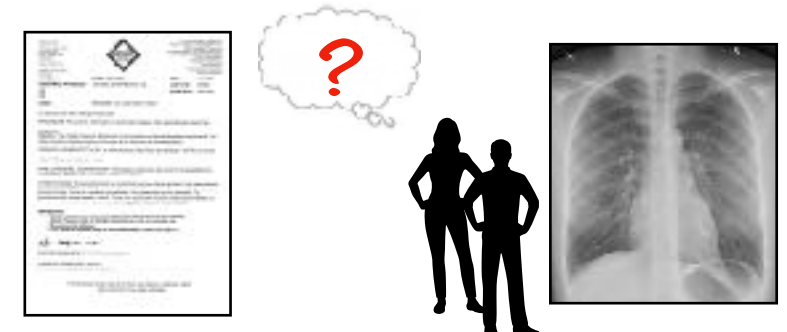
- *Data Used*
- *Image Assessment*
- *Data Exploration*
- *Collaborative Work*

Main outcome:

- **Radiology workflow insights**
- **Key images** should be **linked** to **findings**



Literature review



Poster #1: Visual Storytelling of Big Imaging Data

- Poster presented at *NLeSC 2017 conference*
- “*Interactive*” poster to recall main concept of our project
- *Positive response* of the audience



university of
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Visual Storytelling of
Big Imaging Data

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Project Description

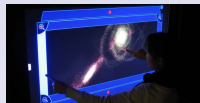
The main research question in this project is how to develop IT support for diagnostic and decision-making processes based on large and complex imaging data. The approach is based on developing novel graphics, visualization, and interaction methods for the exploration of imaging data and the use of storytelling as a means of visual data communication is a key element of the project.

Big Data Visualization

To help scientists understand and explore data through visual representation, intuitive visualization methods that can easily reveal relevant structures and improve the interactive procedure of filtering and analysis will be developed and used. In view of our primary use case (radiology) where data sets increasingly comprise volumetric scans (CT, MRI, PET), we need to include 3D image data in our approach.

Human-Computer Interaction

The analysis of large imaging data includes several types of interaction: analysis, diagnosis and decision making that may lead to follow-up actions. In all these processes, both large-scale and precise interactions with the data are required. Thus, new touch-based techniques have recently been investigated for this purpose [1].




3D visualization exploration widgets that allow users to control the view in 7 degrees of freedom.

Visual Storytelling

Visual storytelling is not only a good way to present data, but also an effective way for another person to pick up the findings and information which is crucial in collaborative and decision-making settings such as healthcare. Storytelling may include several kinds of information and ties these into a visual story, creating a connection between the images and the conclusions to be drawn from the data [2]. Using the CLUE model, an initial story can be edited through *authoring*, creating different branches and a back-link from the *presentation* to the *exploration* stage, which allows users to reproduce the findings and to make new discoveries [3].



The diagnosis process in healthcare based on the traditional workflow (left) and the visual storytelling approach (right).

References

[1] L. Yu, P. Svetachov, P. Isenberg, M. H. Everts, and T. Isenberg. F3D: Direct-touch interaction for the exploration of 3D scientific visualization spaces. *IEEE Transactions on Visualization and Computer Graphics*, 16(6):1613-1622, 2010.
[2] B. Lee, N.H. Riche, P. Isenberg, S. Carpendale: More Than Telling a Story: Transforming Data into Visually Shared Stories. *IEEE Comp. Graph. and Applications*, 2015.
[3] S. Gratzl, A. Lee, N. Gehlenborg, N. Cosgrove, M. Streit: From Visual Exploration to Storytelling and Back Again. *Comput. Graph. Forum* 35(3): 491-500, 2016.

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Provenance





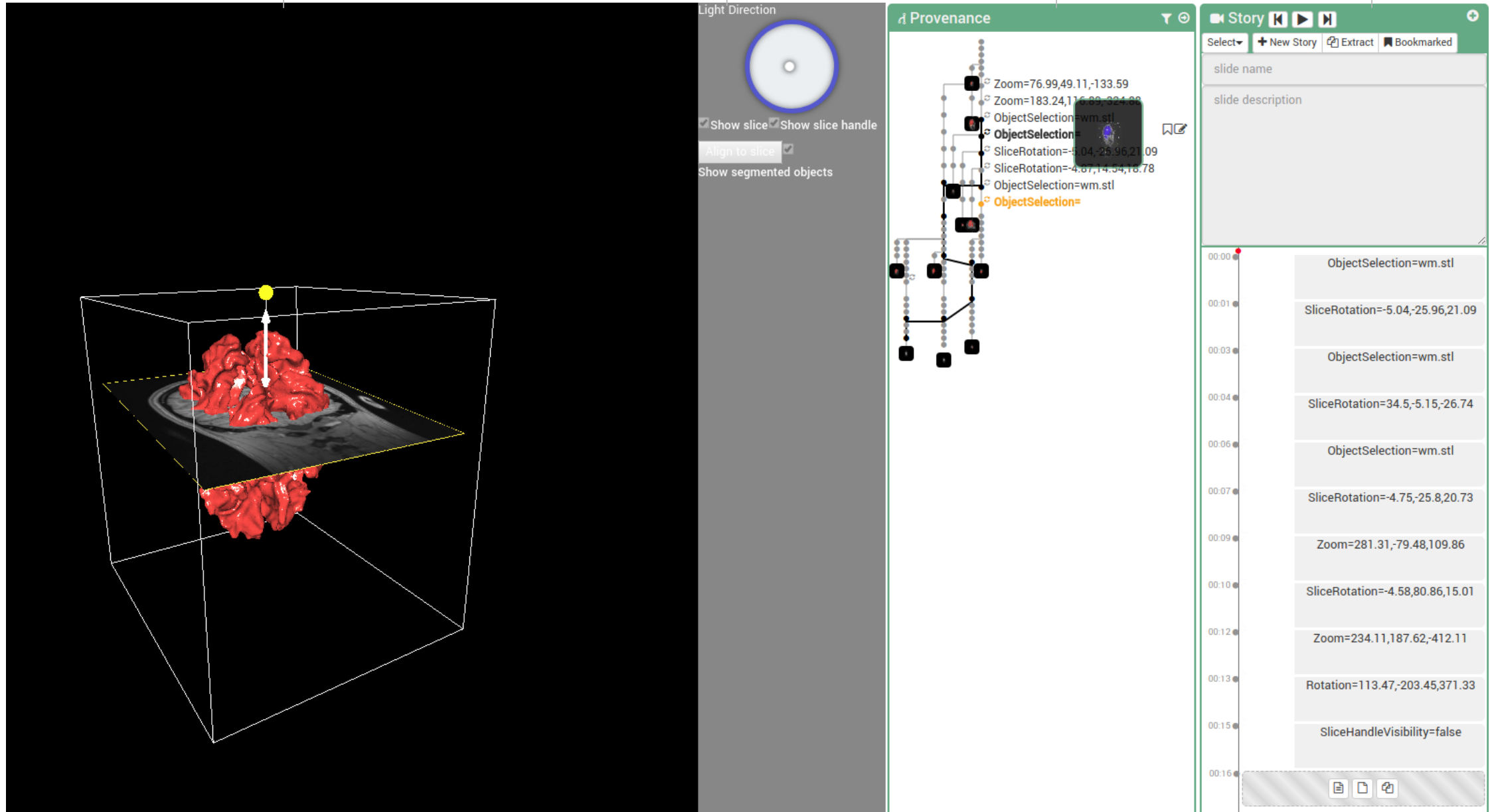
Prototype Development

*Main view
for data exploration*

Configuration settings

Provenance graph

*Authoring tool
for story creation*



The interface is divided into four main sections:

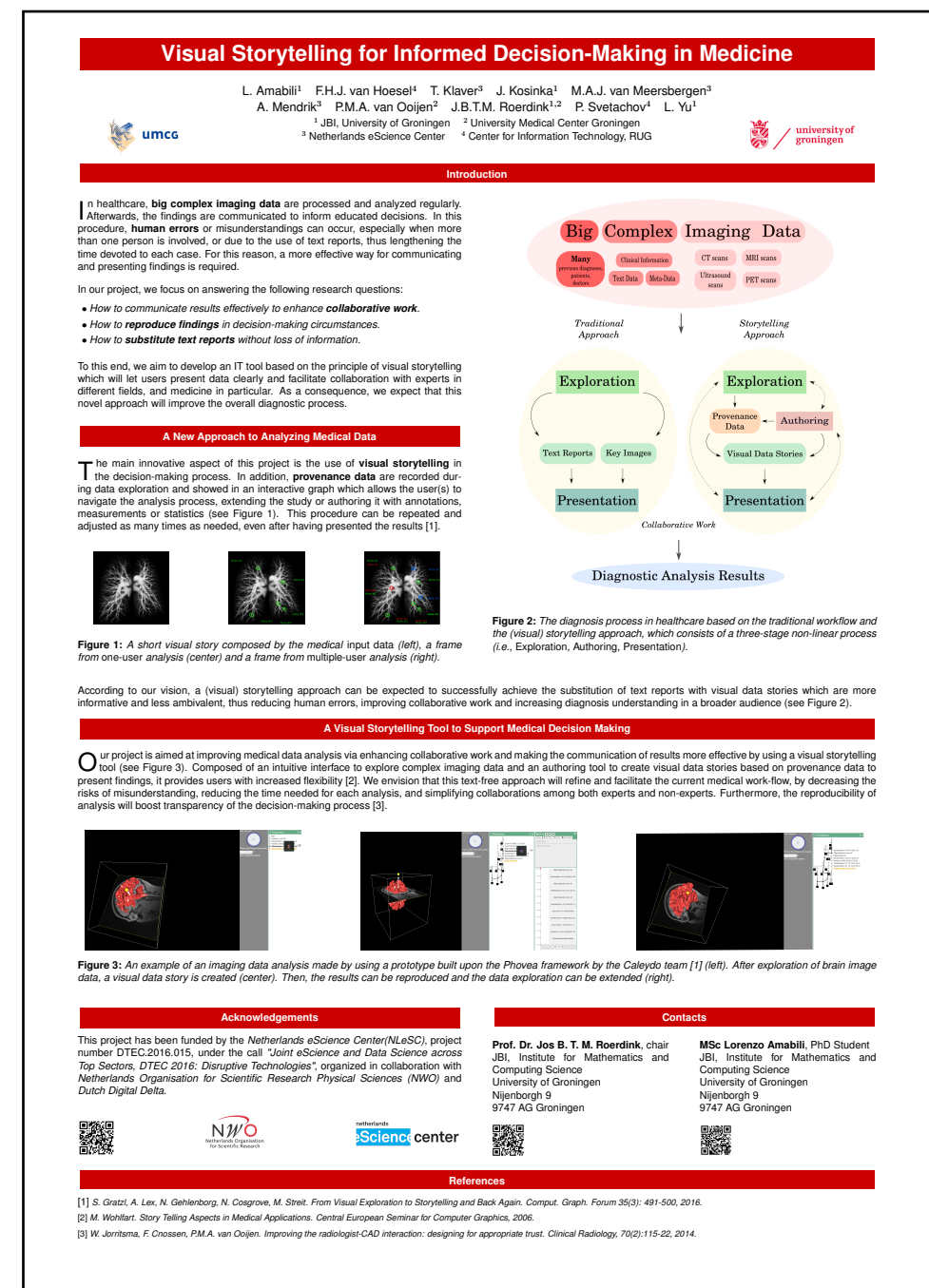
- Main view for data exploration:** A 3D visualization of a red, irregularly shaped object (likely a brain scan) within a wireframe cube. A yellow dot and a white arrow indicate a point of interest on the object's surface.
- Configuration settings:** A panel on the right side of the main view. It includes a 'Light Direction' control (a circular dial), checkboxes for 'Show slice' and 'Show slice handle', an 'Align to slice' checkbox, and a 'Show segmented objects' checkbox.
- Provenance graph:** A panel titled 'Provenance' showing a hierarchical tree of operations. The operations include:
 - Zoom=76.99,49.11,-133.59
 - Zoom=183.24,116.88,-334.88
 - ObjectSelection=wm.stl
 - ObjectSelection=
 - SliceRotation=-5.04,-25.96,21.09
 - SliceRotation=-4.87,14.54,18.78
 - ObjectSelection=wm.stl
 - ObjectSelection=
- Authoring tool for story creation:** A panel titled 'Story' with a timeline and a list of operations. The timeline shows a sequence of operations from 00:00 to 00:16. The operations listed are:
 - ObjectSelection=wm.stl
 - SliceRotation=-5.04,-25.96,21.09
 - ObjectSelection=wm.stl
 - SliceRotation=34.5,-5.15,-26.74
 - ObjectSelection=wm.stl
 - SliceRotation=-4.75,-25.8,20.73
 - Zoom=281.31,-79.48,109.86
 - SliceRotation=-4.58,80.86,15.01
 - Zoom=234.11,187.62,-412.11
 - Rotation=113.47,-203.45,371.33
 - SliceHandleVisibility=false

Poster #2: Visual Storytelling for Informed Decision-Making

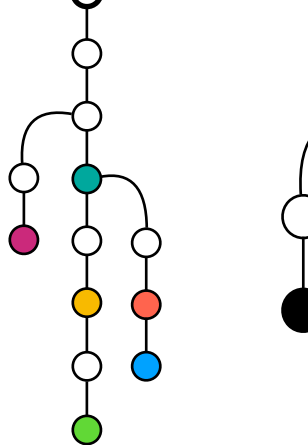
- Poster presented at ICT.OPEN 2018

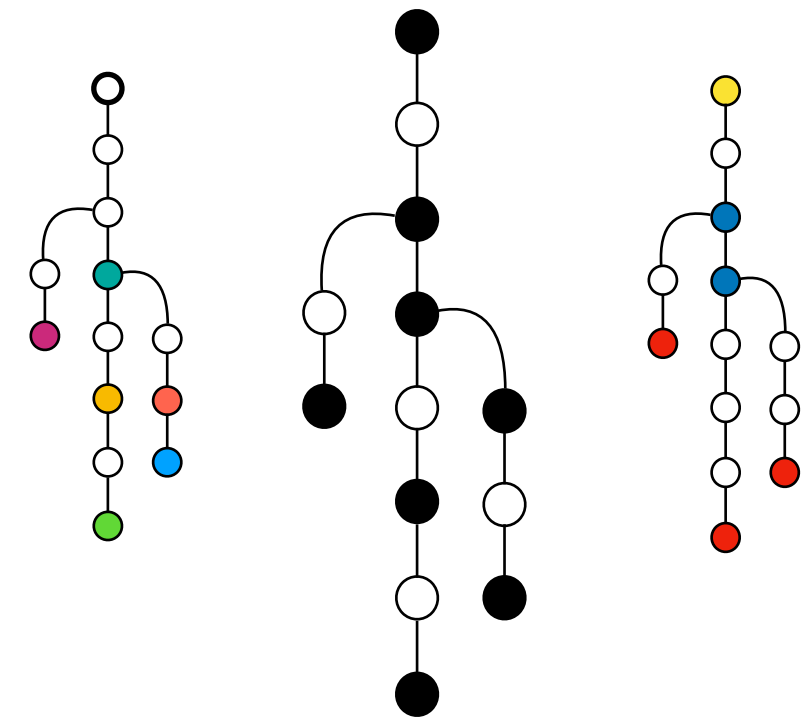
- Project presented to a *broad audience*

- Positive feedback about the potential of the project



***Short paper:* Improving Provenance Data Interaction for Visual Storytelling in Medical Imaging Data Exploration**

- Short paper presented at EuroVis 2018
 - *Positive feedback* about the short paper and the project
 - Two main concepts illustrated:
 - *Prototype development* and its features
 - New algorithm for data aggregation: the *grouping* function
- 
- A diagram illustrating a hierarchical tree structure. The tree has a root node (white) at the top. It branches into two children: a left child (white) and a right child (teal). The left child has a single child (pink). The right child has two children: a left child (white) and a right child (white). The left child of the right child has a single child (yellow). The right child of the right child has a single child (red). The yellow node has a single child (white). The red node has a single child (blue). The white child of the yellow node has a single child (green). To the right of the main tree, there is a separate black node connected to a white node above it.



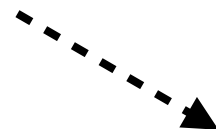
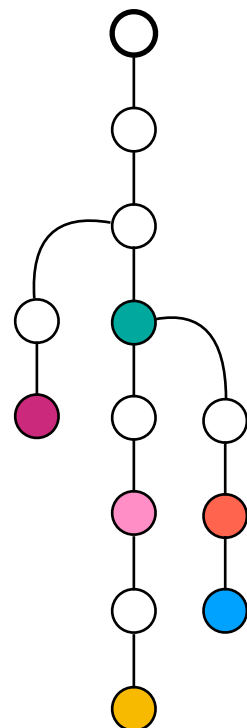
Grouping Feature

○ = non-key node

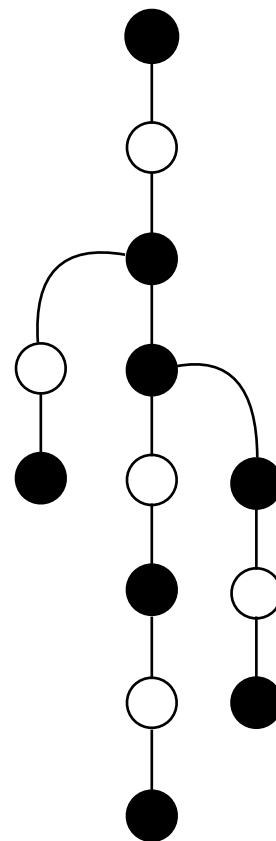
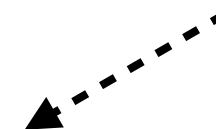
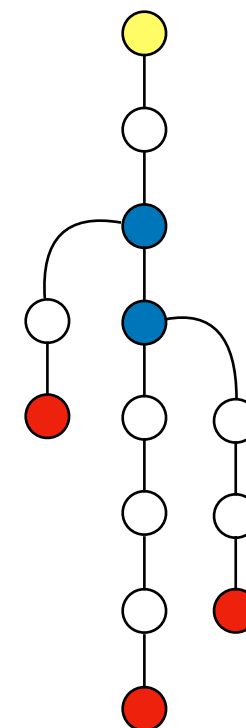
based on *key nodes*

● = key node

Node classification



Topology preservation



- New *rearrangement of the code* and creation of specific libraries
- Brainstorming and planning for the NLeSC, RUG and JKU *future collaboration*
- Discussion of *new research ideas*



Future work

- From Grouping to Automatic Data Aggregation
- Templates for Provenance-based Data Exploration and Visual Data Stories Creation
- Comparison of Provenance Graphs



Caution
Work in progress